# SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Huston Tillotson University
Software Engineering 1

**PROJECT TITLE:** Study Mate – Campus Study Group Finder
**PROJECT DEVELOPER:** Aicha Musfaou
**PROJECT REPOSITORY:** https://github.com/0169-ui

**VERSION;**

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# Revision History

| Name | Date | Reason for Changes | Version |
| --- | --- | --- | --- |
| Aicha Musfaou | Dec 2025 | Initial SRS Draft for Study Mate System | 1.0 |

# 1. INTRODUCTION

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the features, goals, and requirements of the Study Mate platform. The system is designed to help students easily connect with classmates, discover study groups, and organize study sessions in a simple and efficient way.
This document provides a clear reference for developers, testers, and project reviewers throughout the development process.

## 1.2 Document Conventions

- Numbered headings and subheadings are used for organization.
- Requirement priority levels: High (H), Medium (M), Low (L).
- Standard SRS structure from class guidelines is followed.

## 1.3 Intended Audience and Reading Suggestions

- **Developers:** Sections 2–4 contain system details and requirements.
- **Testers:** Section 3 outlines all functional requirements.
- **Instructors/Reviewers:** Sections 1–2 explain the purpose, background, and scope.

## 1.4 Project Scope

Study Mate is a web-based application that helps students find study groups based on their courses, schedules, and academic needs. The platform allows users to search for groups, create new groups, communicate with members, and schedule study sessions.
The main goal of the project is to make collaboration easier and improve the learning experience for students on campus.

## Goals:

- Connect students who share the same classes or study goals.
- Provide tools for organizing and scheduling group study sessions.
- Recommend campus study spaces and resources.
- Support secure login through verified campus emails.

## Benefits:

- Helps students stay organized and build better study habits.
- Encourages teamwork and peer learning.
- Makes it easier to plan and manage group sessions.

## 1.5 References

- University Course Catalog
- Class SRS Template
- IEEE SRS Documentation Guidelines
- Web Accessibility Standards

# 2. OVERALL DESCRIPTION

## 2.1 Product Perspective

Study Mate is a standalone web application created to support academic collaboration on campus. It functions as a central space where students can find classmates, create study groups, and coordinate study sessions.
The system may connect with campus email services for authentication and can optionally integrate with campus facilities data to suggest available study spaces.

## 2.2 Product Features

Key features of the Study Mate platform include:

- Creating and managing study groups
- Searching groups by course code or subject
- Real-time group messaging
- Scheduling tools for availability matching
- Study space recommendations
- Group rating and review system
- Notifications for upcoming sessions

## 2.3 User Classes and Characteristics

**College Students:**
Can search for groups, join groups, message members, and participate in scheduled sessions. Technical skill: Low–Medium.

**Group Organizers:**
Can create groups, update group details, manage membership, and schedule meetings. Technical skill: Medium.

**Campus Administrators (Optional):**
Monitor group activity, address misuse, and manage system permissions. Technical skill: High.

## 2.4 Operating Environment

- Runs on modern browsers: Chrome, Safari, Firefox, Edge
- Supports both mobile and desktop access
- Backend hosted on a cloud server with a connected database
- Requires stable internet connection for full functionality

## 2.5 Design and Implementation Constraints

- System must verify campus email addresses for security
- Must follow university privacy rules and data protection policies
- Limited access for users who are not authenticated
- Must operate within standard browser capabilities—no special hardware required

## 2.6 User Documentation

Study Mate includes:

- Online help center
- Frequently Asked Questions (FAQ) page
- Onboarding guide for new users
- Built-in tooltips to explain features

## 2.7 Assumptions and Dependencies

- Students will have active campus emails
- University provides updated course catalog information
- Internet connection is required for all group and messaging features
- Optional integrations depend on available campus APIs

# 3. SYSTEM FEATURES

## 3.1 User Registration and Login

**Description:**
Allows students to create accounts and access the platform securely using their campus email.

**Priority:** High

**Functional Requirements:**

- The system shall require verification through a valid campus email.
- The system shall store and update user profile information.
- The system shall authenticate returning users.

## 3.2 Course-Based Study Group Discovery

**Description:**
Enables students to search for study groups using course codes, professors, or keywords.

**Priority:** High

**Functional Requirements:**

- The system shall allow users to search groups by course code or subject.
- The system shall support keyword-based search.
- The system shall sort search results by date, popularity, or rating.

## 3.3 Create and Manage Study Groups

**Description:**
Allows users to create new study groups and manage existing ones by editing details such as rules, capacity, and meeting frequency.

**Priority:** High

**Functional Requirements:**

- The system shall allow users to create new study groups.
- The system shall enforce group capacity limits.
- The system shall allow group organizers to edit group information.

## 3.4 Group Messaging

**Description:**
Provides a communication space where group members can exchange messages and coordinate activities.

**Priority:** Medium

**Functional Requirements:**

- The system shall support real-time or near real-time messaging.
- The system shall store group chat history.

## 3.5 Scheduling and Calendar Integration

**Description:**
Helps members coordinate meeting times by allowing them to enter their availability and generating recommended session times.

**Priority:** Medium

**Functional Requirements:**

- The system shall allow users to submit availability.
- The system shall generate suggested meeting times based on member availability.

## 3.6 Location and Room Suggestions

**Description:**
Suggests study locations on campus and allows organizers to attach a meeting place to each session.

**Priority:** Low

**Functional Requirements:**

- The system shall store meeting locations for scheduled sessions.
- The system shall provide a list of recommended campus study spaces.

## 3.7 Rating and Review System

**Description:**
Allows users to rate study groups and leave feedback to help others choose effective study partners.

**Priority:** Low

**Functional Requirements:**

- The system shall allow 1–5 star ratings after study sessions.
- The system shall calculate and display average group ratings.

## 3.8 Notification System

**Description:**
Sends alerts and reminders about upcoming meetings, group updates, or changes in schedule.

**Priority:** High

**Functional Requirements:**

- The system shall notify users of new sessions or updates to existing sessions.
- The system shall send reminders before scheduled study sessions.

# 4. EXTERNAL INTERFACE REQUIREMENTS

## 4.1 User Interfaces

The Study Mate platform will provide a simple and intuitive web interface that supports:

- A search bar for finding study groups
- A clean list view for browsing groups
- A messaging interface for group conversations
- A calendar or schedule view for planning study sessions
- Mobile-friendly layouts for smartphones and tablets

All screens will follow responsive design standards for accessibility and usability.

## 4.2 Hardware Interfaces

Study Mate requires only basic hardware:

- Any laptop, desktop, tablet, or smartphone with a web browser
- No special hardware or additional equipment required

## 4.3 Software Interfaces

The platform will integrate with:

- Campus email system for user authentication
- Optional Google Calendar API for session scheduling
- University course catalog data for accurate course listings

## 4.4 Communications Interfaces

- All communication between users and the server shall occur over secure HTTPS connections.
- Real-time features such as messaging may use WebSocket or similar technologies.

# 5. OTHER NONFUNCTIONAL REQUIREMENTS

## 5.1 Performance Requirements

- The system should load search results within 2 seconds under normal campus network conditions.
- Messaging should update in real-time or near real-time.
- The platform should support a large number of active study groups without performance loss.

## 5.2 Safety Requirements

- The system shall protect user data through regular backups.
- The system shall provide proper error messages when issues occur instead of exposing system details.
- In case of server failure, user data should remain safe and recoverable.

## 5.3 Security Requirements

- All passwords must be securely encrypted before storage.
- Users must verify their accounts through campus email before access.
- The system shall apply rate-limiting to prevent spam or abuse.
- Group messaging shall be monitored or flagged for inappropriate content.

## 5.4 Software Quality Attributes

- **Usability:** Interface must be simple and intuitive for students.
- **Reliability:** System should maintain stable performance during peak usage times (e.g., exam weeks).
- **Maintainability:** Codebase should allow easy updates or bug fixes.
- **Scalability:** System should handle growth in users, messages, and groups.

## 6. OTHER REQUIREMENTS

- The system must comply with university data privacy and security policies.
- The system should allow exporting basic group information if required by campus administration.
- Additional requirements may be added as the university updates its digital policies or academic guidelines.

# Appendix A: Glossary

**Study Group:**
A group of students who meet to study or review materials for a specific course.

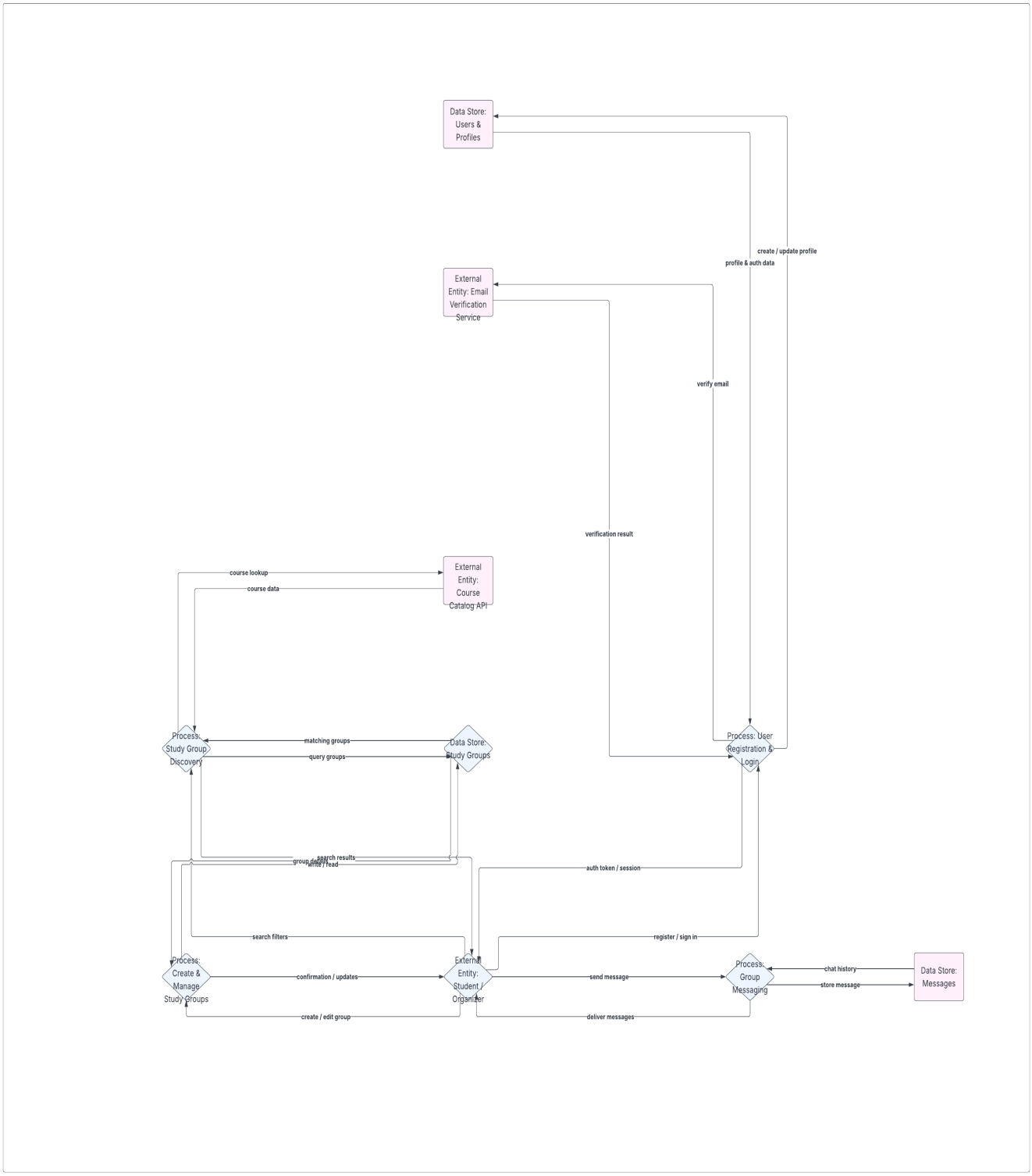
**Organizer:**
A user who creates and manages a study group.

**Availability:**
Time slots submitted by users for scheduling study sessions.

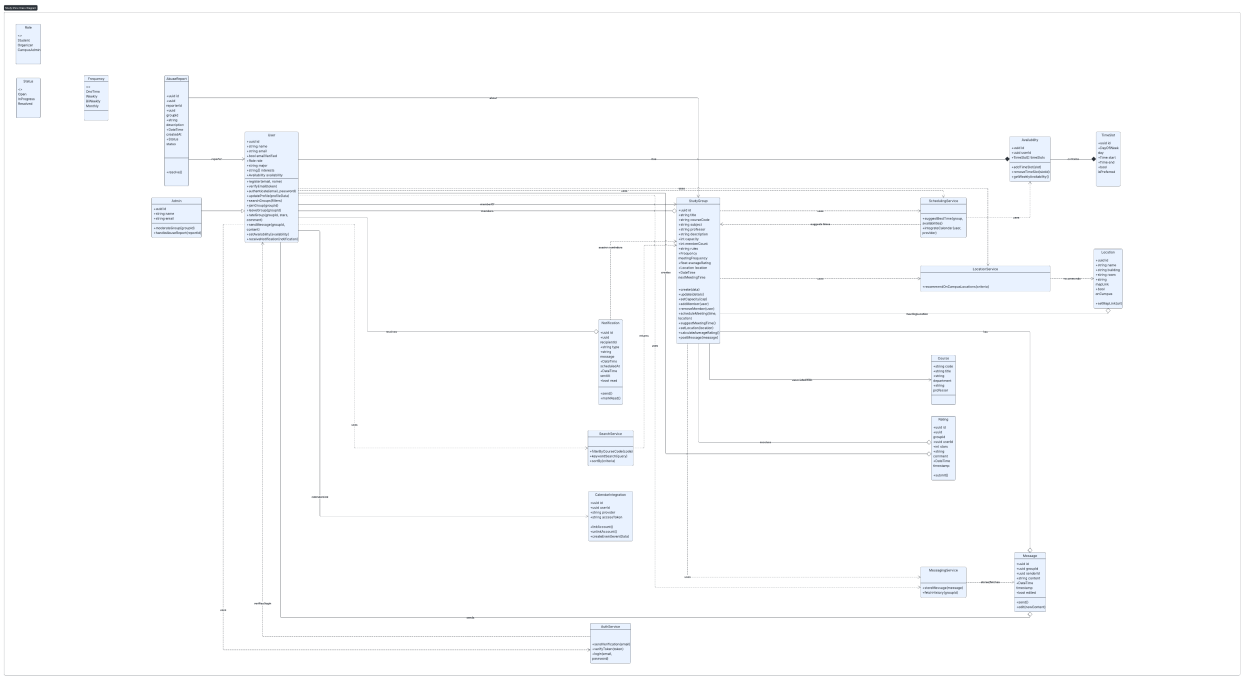
**Rating:**
A 1–5 star value used to evaluate the quality or helpfulness of a study group.

# DATA FLOW DIAGRAM

Study Mate Platform Data Flow Diagram



# CLASS DIAGRAM


Class Diagram
Role
+ +
Student
Organizer
Campusadmin
Status
+ +
Open
InProgress
Resolved
Frequency
+ +
One-Time
Weekly
Biweekly
Monthly
AttendeePasses
+uuid id
+uuid
+uuid eventId
+uuid
groupId
+string
qrCode
+DateTime
checkIn
+DateTime
expiry
+void void()
User
+uuid id
+string name
+string email
+enum emailVerified
+string
+string image
+string[] interests
+Availability availability
+enum[] tutorial_roles
+enum[] publication_roles
+enum[] attendeeRoles[] ,resource[]
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+uint[] preferredTimes
+uint[] notificationsId
+uint[] studyGroupsId
+uint[] ownedGroupsId, int[] ,
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+uint[] ownedResourcesgroupsId,
+uint[] ownedCourses
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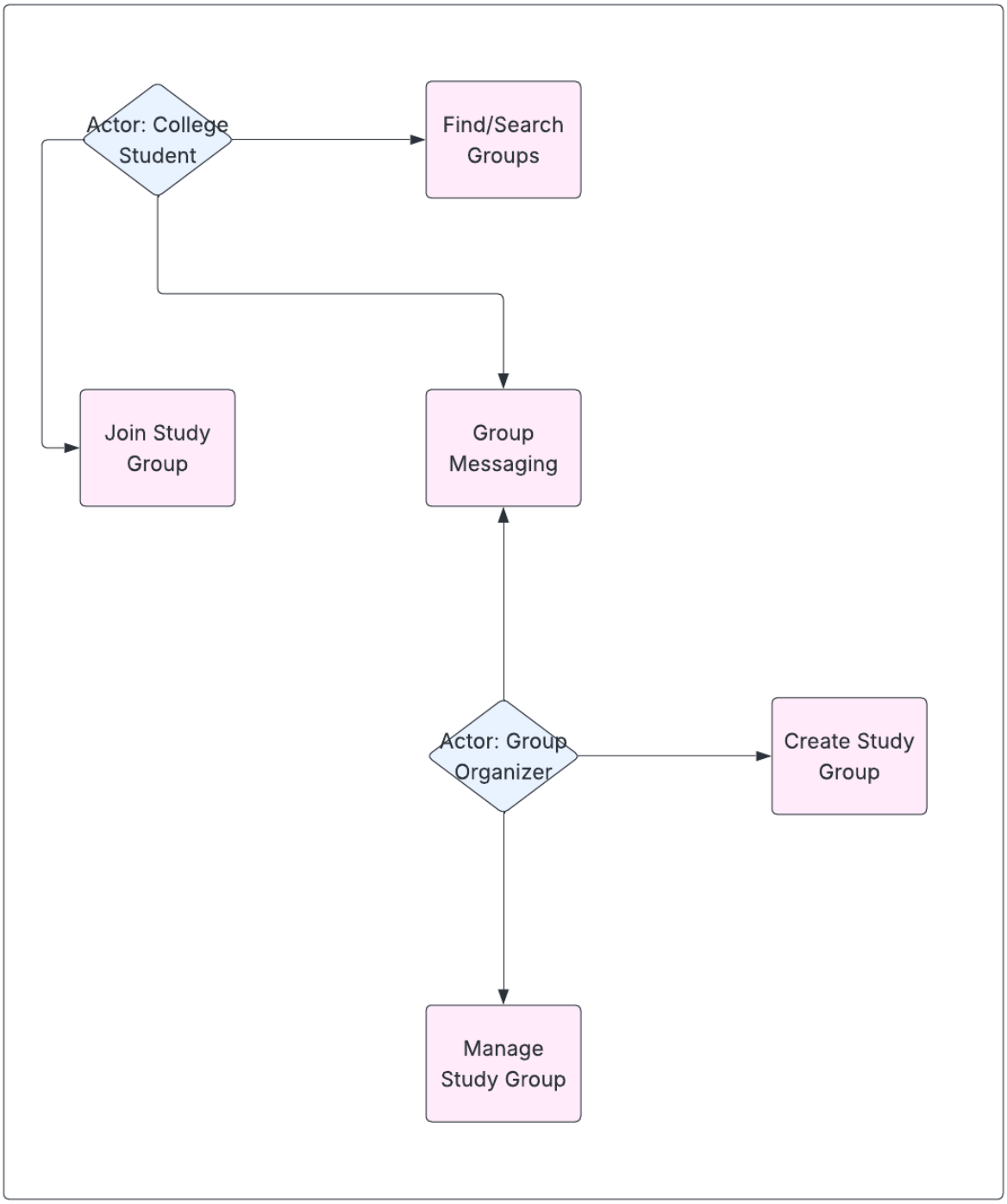
Study Mate Use Case Diagram